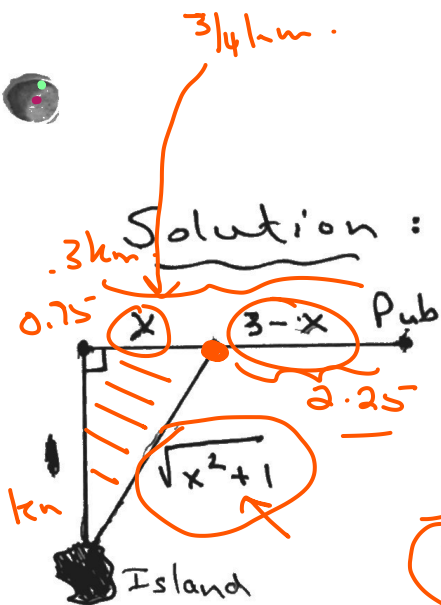
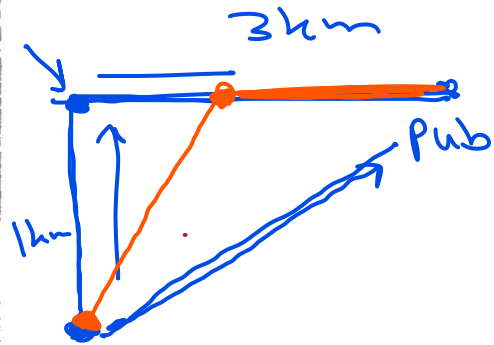


Best Time / Shortest Distance Problems

(1)

Example 1: A man lives on an island 1 km from the shoreline. His favourite pub is 3 km along the shoreline from the point on the shore closest to the island. The man can paddle his canoe at 3 km/hr and can jog 5 km/hr. Determine where he should land in order to reach the pub in the shortest possible time.



Let x be the distance from the point on the shoreline closest to the island.

$$S = \frac{D}{T}$$

$$\Rightarrow T = \frac{D}{S}$$

$$\text{Time} = \frac{\text{Distance}}{\text{Speed}}$$

$$\text{Total Time} = T_{\text{paddle}} + T_{\text{jog}}$$

$$\text{Total Time} = \frac{\sqrt{x^2 + 1}}{3} + \frac{3 - x}{5}$$



(2)

$$\left(\frac{\text{Total}}{\text{Time}} \right)' = \frac{1}{3} \cdot \frac{1}{2} \cdot (x^2+1)^{-\frac{1}{2}} \cdot (2x) - \frac{1}{5}$$

$$\frac{2x}{6\sqrt{x^2+1}} - \frac{1}{5} = 0$$

$$\frac{x}{3\sqrt{x^2+1}} = \frac{1}{5}$$

$$\boxed{\frac{x}{\sqrt{x^2+1}} = \frac{3}{5}}$$

$$\frac{x^2}{x^2+1} = \frac{9}{25}$$

$$25x^2 = 9x^2 + 9$$

$$16x^2 = 9$$

$$x^2 = \frac{9}{16}$$

$$x = \pm \frac{3}{4} \quad (\text{reject } -\frac{3}{4})$$

$\therefore$ He should paddle to a point $\frac{3}{4}$ km from the point closest to the island,

(OR)

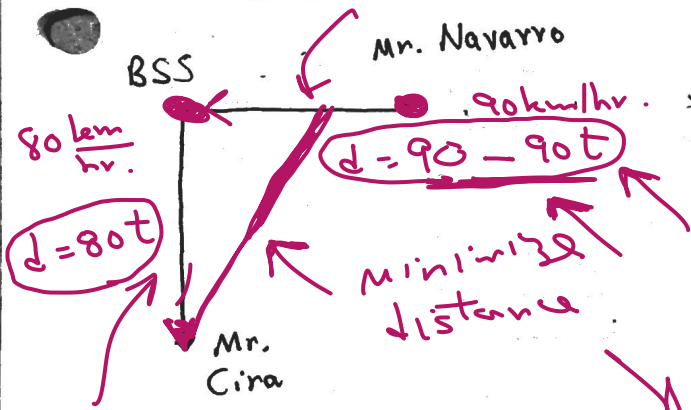
$\therefore$ He should paddle to a point $\frac{9}{4}$ km from the pub.

Example 2: Mr. Cira leaves Bayview S.S. at 1:00 pm and drives due south at 80 km/hr. ^{At same time.} Mr. Navarro is heading due west at 90 km/hr and arrives at Bayview S.S. at 2:00 pm. Determine when they were closest together.

$$s = \frac{d}{t} \Rightarrow d = s \cdot t$$

Solution:

Distance = Speed x Time.



* Mr. Cira's Distance from Bayview S.S. = $80t$

* Mr. Navarro's Distance from Bayview S.S. = $90(1-t)$
 $= 90 - 90t$

$$\therefore d = \sqrt{(80t)^2 + [90(1-t)]^2}$$

$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

1 pm 2 pm

$$d' = \frac{12,800t - 16,200 + 16,200t}{-2\sqrt{(80t)^2 + [90(1-t)]^2}} = 0$$

0.56×60

$$d' = 0$$

$$\therefore 29,000t - 16,200 = 0$$

$$t = 0.559 \text{ hrs } (\odot) \approx 1:33 \text{ pm.}$$